

Detailed Study Guide: Multi-Armed Bandit Quiz Review

ε -Greedy, Sample Averages, Optimistic Initialization, UCB, and Associative Search

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How to Use This Guide

This guide rewrites each quiz question as a study problem and gives the full solution path. For multiple-choice questions, only the correct reasoning is developed; incorrect choices are not analyzed. The emphasis is on how to reproduce the answer from first principles.

Core Ideas to Remember

ε -Greedy Action Selection

In ε -greedy action selection:

- With probability $1 - \varepsilon$, the algorithm exploits by choosing a greedy action.
- With probability ε , the algorithm explores by choosing randomly among all k actions.

If there is a unique greedy action, the probability that the greedy action is selected is

$$P(\text{greedy action}) = (1 - \varepsilon) + \varepsilon \frac{1}{k}.$$

The subtle point is that the random exploration case can still select the greedy action.

Sample-Average Action-Value Estimate

When the action-value estimate uses sample averages,

$$Q_t(a) = \frac{\text{sum of rewards received after selecting action } a}{N_t(a)},$$

where $N_t(a)$ is the number of times action a has been selected before time t . The greedy action before time t is any action with the largest current estimate $Q_t(a)$.

Upper-Confidence-Bound Action Selection

UCB selects

$$A_t = \arg \max_a \left[Q_t(a) + c \sqrt{\frac{\ln t}{N_t(a)}} \right].$$

The term $Q_t(a)$ favors actions with high estimated reward. The bonus term

$$c \sqrt{\frac{\ln t}{N_t(a)}}$$

favors actions with high uncertainty. If an action has been selected fewer times, then $N_t(a)$ is smaller and the uncertainty bonus is larger.

Associative Search

In an ordinary bandit task, the learner chooses actions without observing a state or context. In an associative search task, the learner is told some context, such as whether the current case is A or B. The optimal behavior can therefore depend on the observed case.

1 Question 1: Probability of Selecting the Greedy Action

Problem

In ε -greedy action selection, for the case of two actions and $\varepsilon = 0.5$, what is the probability that the greedy action is selected?

Detailed solution

There are two mutually exclusive ways the greedy action can be selected.

First, exploitation occurs with probability $1 - \varepsilon$. In the exploitation case, the greedy action is selected directly.

Second, exploration occurs with probability ε . In the exploration case, the algorithm chooses uniformly at random among all actions. Since there are two actions, the probability that this random choice happens to pick the greedy action is $1/2$.

Therefore,

$$P(\text{greedy action}) = P(\text{exploit}) + P(\text{explore and randomly pick greedy}).$$

So

$$P(\text{greedy action}) = (1 - \varepsilon) + \varepsilon \cdot \frac{1}{2}.$$

Substitute $\varepsilon = 0.5$:

$$P(\text{greedy action}) = (1 - 0.5) + 0.5 \cdot \frac{1}{2}.$$

Compute each term:

$$1 - 0.5 = 0.5,$$

$$0.5 \cdot \frac{1}{2} = 0.25.$$

Thus,

$$P(\text{greedy action}) = 0.5 + 0.25 = 0.75.$$

Answer

0.75

Study note

Do not answer $1 - \varepsilon = 0.5$. That ignores the fact that the random exploration case can still pick the greedy action.

2 Question 2: Time Steps on Which the ε Case Could Have Occurred

Problem

Consider a k -armed bandit problem with $k = 4$ actions, denoted 1, 2, 3, 4. The algorithm uses ε -greedy action selection, sample-average action-value estimates, and initial estimates

$$Q_1(a) = 0 \quad \text{for all } a.$$

The initial sequence of actions and rewards is

$$A_1 = 1, R_1 = -1,$$

$$A_2 = 2, R_2 = 1,$$

$$A_3 = 2, R_3 = -2,$$

$$A_4 = 2, R_4 = 2,$$

$$A_5 = 3, R_5 = 0.$$

On which time steps could the ε case possibly have occurred?

Key idea

The ε case means the algorithm selected an action randomly among all four actions. Therefore, any selected action could have come from the ε case, even if the selected action was greedy. A selected non-greedy action definitely implies the ε case, but a selected greedy action could still possibly have been caused by random exploration.

Detailed solution

We track the action-value estimates before each action is selected.

Before time step 1

Initially,

$$Q_1(1) = Q_1(2) = Q_1(3) = Q_1(4) = 0.$$

All four actions are tied for greedy. The selected action is $A_1 = 1$. Since action 1 is greedy, this could have been exploitation. But because random exploration can also select action 1, it could also have been the ε case.

After receiving $R_1 = -1$, the estimate for action 1 becomes

$$Q(1) = -1.$$

The other actions remain at 0.

Before time step 2

The estimates are

$$Q(1) = -1, \quad Q(2) = Q(3) = Q(4) = 0.$$

The greedy actions are 2, 3, 4. The selected action is $A_2 = 2$. Since action 2 is greedy, this could have been exploitation. It could also have been the ε case.

After receiving $R_2 = 1$, the estimate for action 2 is

$$Q(2) = 1.$$

Before time step 3

The estimates are

$$Q(1) = -1, \quad Q(2) = 1, \quad Q(3) = 0, \quad Q(4) = 0.$$

The unique greedy action is action 2. The selected action is $A_3 = 2$. This could have been exploitation. It could also have been random exploration selecting action 2.

After receiving $R_3 = -2$, action 2 has rewards 1 and -2 , so its sample-average estimate becomes

$$Q(2) = \frac{1 + (-2)}{2} = \frac{-1}{2} = -0.5.$$

Before time step 4

The estimates are

$$Q(1) = -1, \quad Q(2) = -0.5, \quad Q(3) = 0, \quad Q(4) = 0.$$

The greedy actions are 3 and 4. The selected action is $A_4 = 2$, which is not greedy. Therefore, this step must have been the ε case. Since it must have occurred, it also could have occurred.

After receiving $R_4 = 2$, action 2 has rewards 1, -2 , 2, so

$$Q(2) = \frac{1 + (-2) + 2}{3} = \frac{1}{3}.$$

Before time step 5

The estimates are

$$Q(1) = -1, \quad Q(2) = \frac{1}{3}, \quad Q(3) = 0, \quad Q(4) = 0.$$

The unique greedy action is action 2. The selected action is $A_5 = 3$, which is not greedy. Therefore, this step must have been the ε case. Since it must have occurred, it also could have occurred.

Summary table

Time	Estimates before action	Greedy action(s)	Selected	Could be ε case?
1	0, 0, 0, 0	1, 2, 3, 4	1	Yes
2	-1, 0, 0, 0	2, 3, 4	2	Yes
3	-1, 1, 0, 0	2	2	Yes
4	-1, -0.5, 0, 0	3, 4	2	Yes
5	-1, 1/3, 0, 0	2	3	Yes

Answer

1, 2, 3, 4, 5

3 Question 3: Time Steps on Which the ε Case Definitely Occurred

Problem

Using the same 4-armed bandit sequence as Question 2, on which time step(s) did the ε case definitely occur?

Detailed solution

A step definitely came from the ε case if the selected action was not greedy. This is because the non- ε case is exploitation, and exploitation always chooses a greedy action.

From Question 2, the greedy actions before each selection were:

Time	Greedy action(s)	Selected action	Definitely ε case?
1	1, 2, 3, 4	1	No
2	2, 3, 4	2	No
3	2	2	No
4	3, 4	2	Yes
5	2	3	Yes

At time step 4, the selected action was 2, but the greedy actions were 3 and 4. Therefore, action 2 could only have been selected through random exploration.

At time step 5, the selected action was 3, but the greedy action was 2. Therefore, action 3 could only have been selected through random exploration.

Answer

4, 5

Study note

“Could have occurred” and “definitely occurred” are different. If the selected action is greedy, the ε case is possible but not necessary. If the selected action is not greedy, the ε case is necessary.

4 Question 4: Long-Run Best Method in Figure 2.2

Problem

In the comparison shown in Figure 2.2, which method will perform best in the long run in terms of cumulative reward and probability of selecting the best action?

$$\varepsilon = 0, \quad \varepsilon = 0.01, \quad \varepsilon = 0.1.$$

Detailed solution

Figure 2.2 compares ε -greedy methods on a 10-armed testbed. The important distinction is between short-run performance and long-run performance.

In the short run, a larger ε such as $\varepsilon = 0.1$ is often helpful because the agent explores more aggressively. That helps it discover the optimal action faster.

In the long run, however, the agent has had enough time to form good estimates of action values. Once the optimal action is known, unnecessary exploration becomes a source of lost reward. With ε -greedy, exploration never completely stops as long as $\varepsilon > 0$.

For a 10-armed bandit, once the optimal action is known, the probability of selecting it under ε -greedy is

$$P(\text{optimal}) = (1 - \varepsilon) + \varepsilon \frac{1}{10}.$$

The first term is the exploitation probability. The second term is the probability that exploration occurs and randomly chooses the optimal action from among 10 actions.

Now compare the three methods:

- $\varepsilon = 0$ is purely greedy. It may get stuck with a suboptimal action if early random rewards make that action look best.
- $\varepsilon = 0.1$ explores forever on 10% of steps. This helps early learning but causes persistent long-run loss.
- $\varepsilon = 0.01$ still explores, so it can eventually correct bad early estimates, but it explores only 1% of the time. This gives better asymptotic behavior than $\varepsilon = 0.1$.

Thus, in the long run, the best of the three is the small but nonzero exploration rate, $\varepsilon = 0.01$.

Answer

$\varepsilon = 0.01$

Study note

The phrase “long run” is essential. In the first 1000 steps, $\varepsilon = 0.1$ can look better, but as time goes to infinity, its larger persistent exploration cost becomes worse than $\varepsilon = 0.01$.

5 Question 5: Eventual Difference in Optimal-Action Selection Percentage

Problem

What will be the eventual difference in percentage of optimal action selection between the two long-run methods $\varepsilon = 0.01$ and $\varepsilon = 0.1$ in the 10-armed bandit?

Detailed solution

Assume both methods have eventually learned the true optimal action. The only remaining source of non-optimal action selection is the continuing random exploration.

For $k = 10$ actions,

$$P(\text{optimal}) = (1 - \varepsilon) + \varepsilon \frac{1}{10}.$$

For $\varepsilon = 0.01$:

$$P_{0.01}(\text{optimal}) = (1 - 0.01) + 0.01 \cdot \frac{1}{10}.$$

Simplify:

$$P_{0.01}(\text{optimal}) = 0.99 + 0.001 = 0.991.$$

As a percentage,

$$0.991 = 99.1\%.$$

For $\varepsilon = 0.1$:

$$P_{0.1}(\text{optimal}) = (1 - 0.1) + 0.1 \cdot \frac{1}{10}.$$

Simplify:

$$P_{0.1}(\text{optimal}) = 0.9 + 0.01 = 0.91.$$

As a percentage,

$$0.91 = 91\%.$$

The difference is therefore

$$99.1\% - 91\% = 8.1\%.$$

Answer

$$\boxed{8.1\%}$$

6 Question 6: Early Oscillations and Spikes with Optimistic Initialization

Problem

Figure 2.3 shows the effect of optimistic initial action-value estimates on the 10-armed testbed. The optimistic method uses

$$Q_1(a) = 5, \quad \varepsilon = 0.$$

Both methods use constant step-size parameter $\alpha = 0.1$.

Why are there oscillations and spikes in the early part of the curve for the optimistic method? In other words, what makes this method perform particularly better or worse, on average, on particular early steps?

Detailed solution

The optimistic method is greedy because $\varepsilon = 0$. Normally, a purely greedy method does not explicitly explore. However, optimistic initialization creates an implicit exploration effect.

At the beginning, every action has the same very high estimate:

$$Q_1(a) = 5 \quad \text{for all actions } a.$$

In the 10-armed testbed, true action values and observed rewards are typically much lower than 5. Therefore, when an action is selected once, its estimate is updated downward. With constant step-size update,

$$Q_{t+1}(A_t) = Q_t(A_t) + \alpha(R_t - Q_t(A_t)).$$

Since $Q_t(A_t)$ starts at 5 and R_t is usually much less than 5, the update usually lowers the estimate of the selected action.

After one action has been selected and its estimate has dropped, the remaining untried actions still have their optimistic estimate of 5. Since the method is greedy, it then tends to choose one of those still-untried actions. This causes the algorithm to sweep through the actions early on.

For a 10-armed bandit, this means the first 10 actions are roughly a sweep through all 10 actions in some order. After each action has been tried once, the algorithm has one observed reward from each action. On the 11th step, it tends to choose the action that looked best during this first sweep.

That action is more likely than a randomly selected action to be the true optimal action, so the average percentage of optimal action selection can spike around that time. However, because the value estimates are still heavily influenced by their optimistic starting values and only a few rewards have been observed, subsequent updates can change the ranking of actions. This produces dips and oscillations in the early curve.

Answer

The early oscillations occur because optimistic initialization makes the greedy method try different actions early. In a 10-armed bandit, the first 10 steps tend to sweep through the actions. On the 11th step, the method selects the action that did best in that first sweep, which is more likely than chance to be optimal. But the estimates are still inaccurate and optimistic, so subsequent updates can lower that action's estimate and cause the method to switch, creating dips and oscillations.

Study note

Optimistic initialization is a way to encourage exploration without setting $\varepsilon > 0$. The exploration is caused by overestimating untried actions, not by random exploration.

7 Question 7: UCB Spike on the 11th Step

Problem

In Figure 2.4, the UCB algorithm shows a distinct spike in performance on the 11th step. Why does the reward increase at this step?

Detailed solution

The UCB action-selection rule is

$$A_t = \arg \max_a \left[Q_t(a) + c \sqrt{\frac{\ln t}{N_t(a)}} \right].$$

In the 10-armed bandit case, there are $k = 10$ actions.

At the very beginning, some actions have not yet been tried. For an untried action, $N_t(a) = 0$, so the usual UCB expression is not directly defined because it would require division by zero. In practice, UCB handles this by ensuring that each action is tried at least once before applying the normal formula to all actions.

Therefore, during the first 10 steps, UCB effectively samples each action once. These first 10 actions are not selected because of reliable value estimates; they are selected to remove complete uncertainty about untried actions.

After the first 10 steps, each action has been selected once:

$$N_{11}(a) = 1 \quad \text{for every action } a.$$

At time step 11, the uncertainty bonus is therefore the same for all actions:

$$c \sqrt{\frac{\ln 11}{1}}.$$

Because the bonus is equal across actions, the action with the largest UCB value is the action with the largest current sample-average estimate $Q_{11}(a)$. Since each action has been sampled once, this is the action that received the highest reward in the first 10 samples.

The action that produced the highest reward among the first 10 samples is more likely than a random action to be the true optimal action. Therefore, the average reward increases sharply on the 11th step.

Answer

The reward increases on step 11 because this is the first step after each of the 10 actions has been tried once. At that point, all actions have the same uncertainty count, so UCB chooses the action with the highest observed reward from the first sweep. That action is more likely than chance to be the true optimal action, producing a spike in average reward.

Study note

The 11th-step spike is analogous to choosing the winner of the first round of samples. It is not because UCB already knows the true best action; it only has one noisy sample from each action.

8 Question 8: UCB Reward Decrease After the 11th Step

Problem

Why does the UCB reward decrease on the subsequent steps after the 11th-step spike? Hint: if $c = 1$, the spike is less prominent.

Detailed solution

At step 11, UCB chooses the action that looked best after each of the 10 actions had been tried once. Suppose that action is a^* for the moment. Before step 11, each action had count 1:

$$N_{11}(a) = 1 \quad \text{for all } a.$$

After action a^* is selected on step 11, its count increases to 2:

$$N_{12}(a^*) = 2.$$

The other actions still have count 1:

$$N_{12}(a) = 1 \quad \text{for } a \neq a^*.$$

Now compare the uncertainty bonuses at the next step. For the selected action a^* , the bonus is

$$c\sqrt{\frac{\ln 12}{2}}.$$

For each other action, the bonus is

$$c\sqrt{\frac{\ln 12}{1}}.$$

Since

$$\sqrt{\frac{\ln 12}{2}} < \sqrt{\ln 12},$$

the action selected on step 11 now has a smaller uncertainty bonus than the other actions.

If c is large, this difference in uncertainty bonus is large. In Figure 2.4, $c = 2$, so the penalty for having been selected again is strong. As a result, the action that performed best in the first 10 steps can be temporarily ruled out on step 12, even if it is promising. UCB is pulled back toward less-sampled actions because their uncertainty bonuses are larger.

This causes the reward to decrease after the spike. The spike is less prominent when $c = 1$ because the uncertainty bonus is smaller, so the algorithm is less aggressively pulled away from the action chosen on step 11.

Answer

The reward decreases after step 11 because the action chosen at step 11 has now been selected twice, so its uncertainty bonus becomes smaller than the bonuses of the other actions. With $c = 2$, this uncertainty effect is strong enough to make UCB switch away from that high-performing action on subsequent steps, lowering average reward.

Study note

UCB balances exploitation and uncertainty. The decrease after step 11 is not random exploration. It is the deterministic effect of the uncertainty bonus favoring actions that have been selected fewer times.

9 Question 9: Expected Return Without Knowing the Case

Problem

Suppose you face a 2-armed bandit task whose true action values change randomly from time step to time step. At any time step, the true values of actions 1 and 2 are respectively

0.1 and 0.2

with probability 0.5; call this case A. They are respectively

0.9 and 0.8

with probability 0.5; call this case B.

If you are not able to tell which case you face at any step, what is the expected return you can achieve?

Detailed solution

If the case is hidden, then you cannot condition your action on whether the current step is case A or case B. You must compare the unconditional expected value of each action.

For action 1:

$$\mathbb{E}[q(a_1)] = 0.5 \cdot 0.1 + 0.5 \cdot 0.9.$$

Compute:

$$\mathbb{E}[q(a_1)] = 0.05 + 0.45 = 0.50.$$

For action 2:

$$\mathbb{E}[q(a_2)] = 0.5 \cdot 0.2 + 0.5 \cdot 0.8.$$

Compute:

$$\mathbb{E}[q(a_2)] = 0.10 + 0.40 = 0.50.$$

Both actions have the same unconditional expected value. Therefore, the best expected return you can achieve is 0.5.

Answer

0.5

Study note

Without observing the case, this is an ordinary bandit decision. You can only choose based on the average value of each action across cases.

10 Question 10: Which Arm to Pull Without Knowing the Case

Problem

Which arm should you pull to achieve the expected return from Question 9?

Detailed solution

From Question 9, the unconditional expected value of action 1 is

$$\mathbb{E}[q(a_1)] = 0.5.$$

The unconditional expected value of action 2 is

$$\mathbb{E}[q(a_2)] = 0.5.$$

Since both actions have exactly the same expected return, either action achieves the maximum expected return.

Answer

Either action

Study note

This does not mean the two actions are equal in every case. In case A, action 2 is better. In case B, action 1 is better. They are equal only after averaging over the hidden cases.

11 Question 11: Expected Return When the Case Is Observed

Problem

Now suppose that on each step you are told whether you are facing case A or case B, although you still do not know the true action values. This is an associative search task. What is the expected return you can achieve?

Detailed solution

Now the case is observed. That means the action can depend on the case.

In case A, the action values are

$$q(a_1) = 0.1, \quad q(a_2) = 0.2.$$

The better action in case A is action 2, giving expected reward

$$0.2.$$

In case B, the action values are

$$q(a_1) = 0.9, \quad q(a_2) = 0.8.$$

The better action in case B is action 1, giving expected reward

$$0.9.$$

Each case occurs with probability 0.5. Therefore, if we choose optimally within each case, the expected return is

$$0.5 \cdot 0.2 + 0.5 \cdot 0.9.$$

Compute:

$$0.5 \cdot 0.2 = 0.1,$$

$$0.5 \cdot 0.9 = 0.45.$$

Thus,

$$\mathbb{E}[\text{return}] = 0.1 + 0.45 = 0.55.$$

Answer

$$\boxed{0.55}$$

Study note

The value increases from 0.5 to 0.55 because observing the case lets you use different actions in different contexts.

12 Question 12: How to Behave in the Associative Search Task

Problem

How should you behave to achieve the expected return from Question 11?

Detailed solution

Because the case is observed, the policy can condition on the case.

For case A:

$$q(a_1) = 0.1, \quad q(a_2) = 0.2.$$

Since

$$0.2 > 0.1,$$

choose action 2 in case A.

For case B:

$$q(a_1) = 0.9, \quad q(a_2) = 0.8.$$

Since

$$0.9 > 0.8,$$

choose action 1 in case B.

This policy achieves

$$0.5 \cdot 0.2 + 0.5 \cdot 0.9 = 0.55.$$

Answer

Action 2 in case A, action 1 in case B

Study note

This is the central idea of associative search: the best action is not fixed globally. It depends on the observed situation.

Final Answer Key

Question	Final answer
1	0.75
2	1, 2, 3, 4, 5
3	4, 5
4	$\varepsilon = 0.01$
5	8.1%
6	Optimistic initialization causes an early sweep through actions; after the first sweep, the best-looking action is selected, causing a spike, followed by oscillations as estimates adjust.
7	Step 11 is the first reward-dependent UCB selection after all 10 actions have been tried once; UCB chooses the action with the highest first observed reward.
8	The action selected on step 11 has its count increased, so its uncertainty bonus decreases; with large c , UCB switches to less-sampled actions, causing the reward to drop.
9	0.5
10	Either action
11	0.55
12	Action 2 in case A, action 1 in case B